

## Activity 02

Q.1) Given a vector field  $D = 2\rho^2 \cos(\phi) a_\rho + 6\rho \sin(\phi) a_\phi$ , evaluate both sides of the divergence theorem for a region bounded by surfaces  $\rho = 2$ ,  $\phi = 0$ ,  $\phi = \pi$ ,  $z = 0$ , and  $z = 5$ .

$$\oint_S A \cdot dS = \int_V (\nabla \cdot A) dV$$

$$\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$\bullet D_\rho = 2\rho^2 \cos \phi, D_\phi = 6\rho \sin \phi, D_z = 0$$

$$\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (2\rho^3 \cos \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (6\rho \sin \phi)$$

$$= \frac{1}{\rho} (6\rho^2 \cos \phi) + \frac{1}{\rho} (6\rho \cos \phi)$$

$$= 6\rho \cos \phi + 6 \cos \phi$$

$$= 6 \cos \phi (\rho + 1)$$

$$\text{Solving } \iiint_V (\nabla \cdot \vec{D}) dV$$

$$= \int_{z=0}^5 \int_{\phi=0}^{\pi} \int_{\rho=0}^2 6 \cos \phi (\rho + 1) d\rho d\phi dz$$



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$$\int_{z=0}^5 \int_{\phi=0}^{\pi} 6 \cos \phi \left[ \frac{p^2}{2} + p \right] \Big|_0^2 d\phi d\theta.$$

$$\int_{z=0}^5 \int_{\phi=0}^{\pi} 24 \cos \phi \cdot d\phi d\theta. \quad \Rightarrow \int_{z=0}^5 24 \sin \phi \Big|_0^{\pi} \cdot d\theta$$

$$\int_{z=0}^5 0 \cdot d\theta = \boxed{0}$$



The closed surface:

$\iint_S D \cdot dS$ :-

curved side  $\rho = 2$

$$dS = \rho d\phi dz \text{ with } \rho = 2$$

Parametrization using  $(\phi, z)$ :

$$\begin{aligned} r(\phi, z) &= (x, y, z) \\ &= (2\cos\phi, 2\sin\phi, z) \end{aligned}$$

Tangent Vectors:

$$r_\phi = \frac{\partial r}{\partial \phi} = (-2\sin\phi, 2\cos\phi, 0)$$

$$r_z = \frac{\partial r}{\partial z} = (0, 0, 1)$$

Area Vector:

$$dS = (r_\phi \times r_z) d\phi dz$$

$$r_\phi \times r_z = \begin{bmatrix} i & j & k \\ -2\cos\phi & 2\sin\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= (2\cos\phi, 2\sin\phi, 0) = 2a_\rho$$

$$\therefore dS = 2a_\rho d\phi dz$$

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Field component on this surface:

$$D_\rho (\rho=2, \phi) = 2(2)^2 \cos \phi = 8 \cos \phi$$

flux through curved side:

$$\text{flux} = \int_{z=0}^5 \int_{\phi=0}^{\pi} D \cdot d\mathbf{s}$$

$$= \int_0^5 \int_0^{\pi} D_\rho (\rho=2, \phi) (2\phi \, d\phi \, dz)$$

$$= \int_0^5 \int_0^{\pi} (8 \cos \phi) (2) \cdot d\phi \, dz$$

$$= \int_0^5 \left[ 16 \sin \phi \right]_0^{\pi} \cdot dz$$

$$= \int_0^5 0 \cdot dz = \boxed{0}$$

Both sides are zero. Hence proved.